LESSON PLAN · SCRATCH JUNIOR (K–2) ADVANCED: GAME BUILDERS

# Lesson 11: Test and Fix Your Game

Module 4 | 30 minutes | Testing and debugging | KG–Grade 2 · ages 6–7

**Objective:** Test a game like a player, find bugs in rules and scores, and fix them.

**Success criteria**

- I can test against the goal.
- I can find bugs in score and rules.
- I retest after fixing.

## Key vocabulary

- **tester**: Someone who tries the game to find problems.
- **requirement**: What the game must do.
- **retest**: Try again after fixing.

## You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

## Timed lesson plan (30 min)

| Time | Stage | What happens |
|---|---|---|
| 5 min | **Warm-up (unplugged)** | Rule check (5 min): play a board game with a missing rule; spot it. |
| 5 min | **Watch** | Play the lesson video with captions; pause at each heading and ask a question. |
| 8 min | **Build along** | Fix the games: learners follow the steps. |
| 5 min | **Perform** | Score bug, No win message, Tester feedback |
| 3 min | **Probe** | Shrinking bug |
| 4 min | **Wrap-up** | Exit ticket: Name one bug you fixed today. |

## Unplugged warm-up

Rule check (5 min): play a board game with a missing rule; spot it.

## Teaching the idea

Testing against requirements (the goal) and debugging rule/state errors: missing increments, wrong loop counts, wrong looks blocks, missing win conditions.

**Analogy:** A taste-tester checking a recipe.

**Live demo:** Play a buggy game and think aloud about what's wrong.

## Questions to ask

| Ask | Listen for |
|---|---|
| Does the score match? | Check. |
| What did the tester notice? | Feedback. |
| What do you do after fixing? | Retest. |

## Differentiation

**Support**
Tell children which part (score, loop, message) has the bug.

**Stretch**
Swap games with a partner and test each other's.

## Common misconceptions

| Misconception | How to fix it |
|---|---|
| If it runs, it's right | It must meet every goal. |

## Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

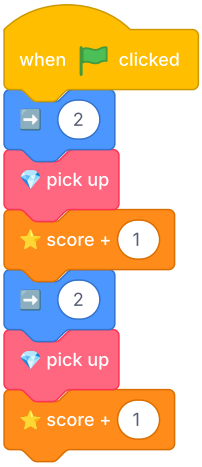
**Exit ticket**
Name one bug you fixed today.

**Homework**
Test a game at home and suggest one improvement.

# Answer key — Kitabash activities

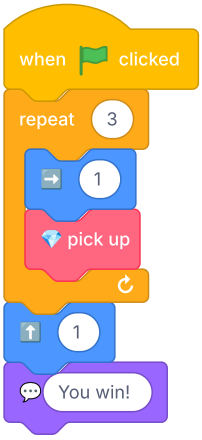
## 1. Score bug (perform · block code)

**One solution:**




## 2. No win message (perform · block code)

**One solution:**

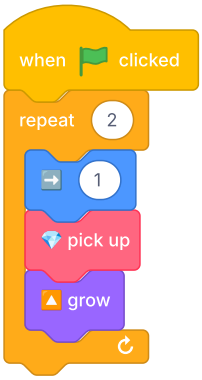



## 3. Tester feedback (perform · multiple choice)

**Answer:** The win message

**4. Shrinking bug** (probe · block code)

**One solution:**




## Answer key — worksheet

1. **Key words:** tester = Someone who tries the game to find problems. · requirement = What the game must do. · retest = Try again after fixing.
2. **Tester feedback:** A. The win message — The game needs to tell the player they won.
3. **Tester checklist:** All ticked.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.